Smart Insole: A Wireless Gait Monitoring and Pressure Distribution Device

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***Abstract*—A wireless gait monitoring and pressure distributing device was successfully fabricated using copper sheet which was attached to a polyethylene terephthalate (PET) film using an adhesive. The fabricated device consisted of sixteen force sensitive resistors (FSR), three ESP32 microcontrollers. Calendering was performed to remove the wrinkles and air trapped in the copper traces to ensure good conductivity followed by lamination of the insole to avoid damage to FSRs, copper and preventing it from oxidizing. Smart insole was responsible for generating pressure and gait data on a computer application and provide the user feedback regarding the pressure applied on their foot. The performance of the device was investigated by measuring the force on each FSR using force gauge between 0 to 150 N of force with 10 N intervals in between the force. It was observed that the device demonstrated swift performance when and promptly sent data to the application while the subject was performing different strides such as heel strike, flat foot, mid stance, push off and, toe off.**

Keywords—Smart Insole, Gait Monitoring, Wireless, Pressure Distribution

I. Introduction

Gait is a measure of a person's ability to walk or run as it involves the coordination and balance of muscles which allow the body to move forward. Instability in gait can lead to various heart related problems [1]. Studies show patients with severe chronic heart failure seem to take shorter steps. There are several factors that can affect the gait movement in humans such as injuring, illness, aging etc. The ability to identify changes in gait movement can help in identifying events such as falls, chronic diseases, posture identification and movement related problems etc. [2-3]. Thus, it is essential to develop a significant gait analysis that can identify and differentiate between normal and pathological diseases. This early identification helps in enabling applications in healthcare, physical therapy, exercises etc. A common example involving importance of gait analysis is to prevent elderly from fall risk or to diagnose progress in Parkinson's disease [4-5].

There are several gait monitoring systems that were investigated upon till date. Generally, gait monitoring is performed in a professionally built gait lab where tests are conducted using different wires being attached to a person's body that help analyze the movements while they are walking on a treadmill [6-7]. During gait study, the subject is supposed on walk by wearing markers that are reflective. The traditional approach involves usage of eight high-speed cameras for capturing joint angles, pressure sensitive carpet in addition to devices such as accelerometers, markers etc. which are body convenient for measuring constant repetitive patterns of foot pressure distribution using pressure sensing walkways [8-26]. Regardless, the conventional approach was very expensive to setup necessary facilities in addition to affecting the gait motion resulting in biased or incorrect gait data [27]. Later several other approaches were researched upon such as using 3-axis accelerometer, sensor array system etc. for gait analysis. However, the accuracy of addressing gait monitoring using these approaches was not extensive increasing the requirement for easy, portable, and professional system for gait analysis [28-31].

To overcome these issues, recently increased attention is paid in the field of wearable devices due to their low production cost and possibility of daily use. Different gait characteristics can be monitored depending on the wearable system developed [32-33]. For instance, inertial measurement system can be useful for determining differences of joints in coronal and sagittal planes. Similarly, joint angle with high sensitivity can be recorded using goniometers. Most of these systems need multiple wires attached to subjects making the gait monitoring complex. Developing wireless smart insole utilizing force sensitive resistors (FSR) where one only needs to apply pressure on them to get proper gait analysis readings are capable of longer uses in everyday life [34-36].

In this work, a wireless smart insole, a gait monitoring and pressure distribution device were fabricated using fifteen force sensitive resistors (FSRs) installed onto copper traces. This virtual soleprint with FSRs helps in attaining precise and accurate pressure readings. Next, copper traces were placed on a polyethylene terephthalate (PET) film which would be placed under the sole of a regular shoe. A circuit board was also fabricated with capability of wirelessly transmitting graphical representation of gait curve and pressure distribution to a computer application. Compared to traditional gait analysis which requires up to fifteen wires for each sensor, wireless smart insole can be accessible anywhere, comfortable, flexible design that can fit any footwear, and provides real time pressure map to the user. Gait analysis was performed on a subject forward motion while walking and gait

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curve analysis was performed depending on different foot positions such as heel strike, flat foot, mid stance, push off, toe off etc.

II. Methodology

A. Copper Trace Design

A flexible printed circuit board (PCB) was developed using copper sheet and PET film. Initially, Autodesk fusion 360 was used to design the shape of the insole with a size of 10.5 male shoe. Later, 3D models of FSR were strategically placed on the insole to get the most accurate gait curve and pressure distribution of the foot. Fig. 1(a) represents the designed Autodesk 360 design concept for sole print with FSRs. Sixteen copper traces were drawn at 1.2 mm width with an offset of 1.3 mm. FSR sensors are a form of variable resistor that changes resistance as force is applied. The higher the pressure, the lower the resistance between the two terminal drops. Fifteen of the copper traces were used for FSR output and the remaining one was used for Vin connected to each FSR. The copper sheet was attached onto the PET film with the help of an adhesive. Later, a calendaring machine was used to eliminate the wrinkles and air trapped in the copper traces which could potentially lead to poor electrical connections. Initially, the calendaring pressure was set to 0 PSI, and some improvement was observed on the wrinkles. Later, the pressure was increased to 5 PSI and 10 PSI to get rid of the air pockets on the copper film print. Increasing the pressure to more than 10 PSI resulted in copper traces being ripped off from the PET. The designed Autodesk file was loaded on to Graphtec plotter machine which was used to cut the insole and copper traces into desired shapes.

After calendering, fifteen FSRs were installed onto the copper tracing using adhesives to ensure that the installed FSRs remained in place. Later, two pins of FSRs were soldered into corresponding copper traces. Later, the copper trace print was laminated to avoid the damage of copper traces when bent or curved. We also noticed that the copper trace would oxidize and start to become a poor conductor if left outside for more than two days. Laminating proved to be the solution for these issues as it protects the copper traces in addition to FSRs and provides a more rigid base. Fig. 1(b-f) represents the Graphtec plotter machine used for fabrication, copper trace attached to PET film using adhesive, calendered copper traces to eliminate wrinkles, copper film trace after peeling off, and smart insole with attached FSRs after lamination.

B. Circuit Flow

The circuit developed for this system involved digital conversion of the analog voltage between the two terminals of FSR, and the processing and wireless transmission of the data to application. One side of the fifteen FSRs was connected to Vcc while the other is connected to the channels of a sixteen-channel analog multiplexer. The signal pin on the multiplexer was connected to a static 3.3 KΩ resistor for voltage divider and to analog input in ESP-32. A 3.3 KΩ resistor was selected since it provided an output that was more linear without consuming too much power. The four

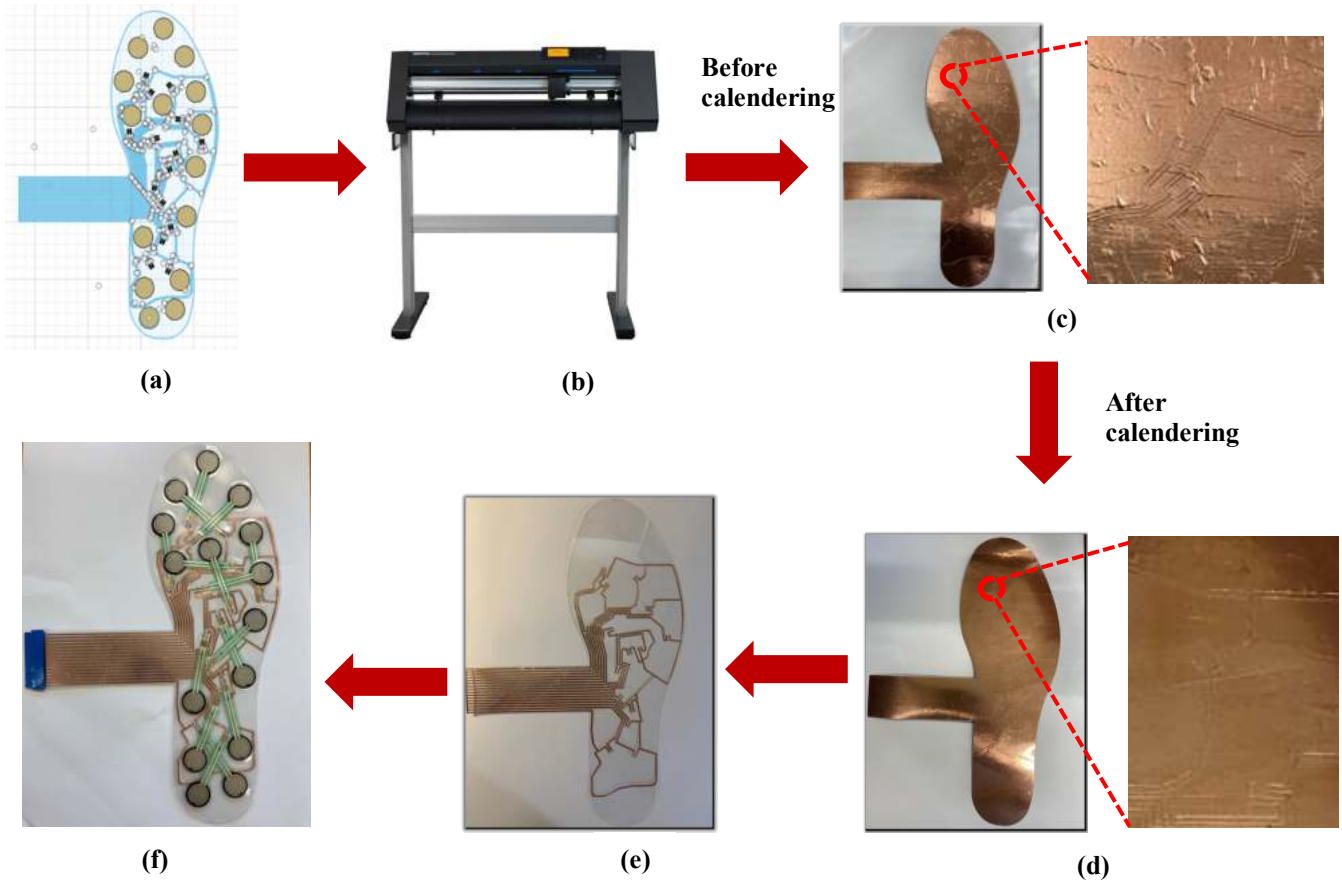

Fig. 1: (a) Designed insole with force sensitive resistors, (b) Graphtec plotter machine used for fabricating the designed insole, (c) copper sheet attached to PET film using an adhesive before calendaring (inset: wrinkles on copper sheet), (d) copper film trace calendaring (inset: no wrinkles observed), (e) peeled off excessive copper trace, and (f) smart insole with attached FSRs after lamination.

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outputs on the MUX are connected to the digital pins in the ESP-32 microcontroller. The data is programmed sequentially to be received alternatively between each FSR with a 1 ms delay.

For processing and wireless transmission of the digital signal, an ESP32-WROOM-32D manufactured by Espressif was used. The module contained a dual core Xtensa 32-bit LX6 MCU which used a 40-MHz crystal oscillator. The module also provides the user with multiple wireless transmission methods (Bluetooth, BLE, & Wi-Fi) and offer advanced power-management technologies. The module can switch between different power modes which is essential for the devices to function properly. Two of the three ESP-32 microcontrollers were installed in the right and left insole while the third acted as the master server that received the data from the other two using Wi-Fi and performed calculations to get the force. The master server was directly connected to the computer for input to a visual studio application. As soon as data was received, it updates the pressure map and the gait curve. This allows quick response as communication between the three microcontrollers is very straightforward and fast.

III. Results & Discussion

A. Deriving Force Equation

After fabricating the insoles, MARK-10force gauge was used to calibrate the sensors. As per the Interlinks datasheet, the sensing range of the FSR 402 UX sensor was between 0.5 to 150 N. Accordingly, we set the force gauge to apply force from 0 to 150 N in time interval of 8 seconds. The produced data was stored in a microcontroller from FSR and was later copied to excel to generate force equation. Fig. 2(a-b) represent the force testing procedure and generated force equation respectively. To generate the force equation, the peaks from the force values were considered and by flipping the axis, we created a new graph which is referred to as the 4th order polynomial equation (Fig. 2(b)).

B. Pressure Map Application and Gait Curve Results

Three ESP32 microcontrollers were used in our circuit. One for right and one for left insole and the last one was the master server that is plugged in the computer for the visual studio application. The left and right microcontrollers wirelessly transmit data to the server board using Wi-Fi. The main serves perform the calculations for pressure after receiving the data. Next a 3D model was designed using Autodesk fusion that holds the battery and microcontroller secure. The designed casing model was 3D printed and the everything was put together inside the casing. Fig. 2(c-f) represent the designed Autodesk casing, 3D printed case, final product where the right insole is connected to the circuit and, smart insole placed inside the shoe for pressure mapping and gait analysis.

Fig. 3 (a-b) represents the layout of the pressure mapping application using windows forms on visual studio with C# and force measured on each sensor respectively. The insole on the left is what was displayed when the left foot is not connected. Each color represents the zones that each FSR sensor covers. Multiple data points were assigned to each FSR to give the pressure map a more visually appealing look. For the application, it was connected by opening whichever COM Port the server device was connected to. After choosing

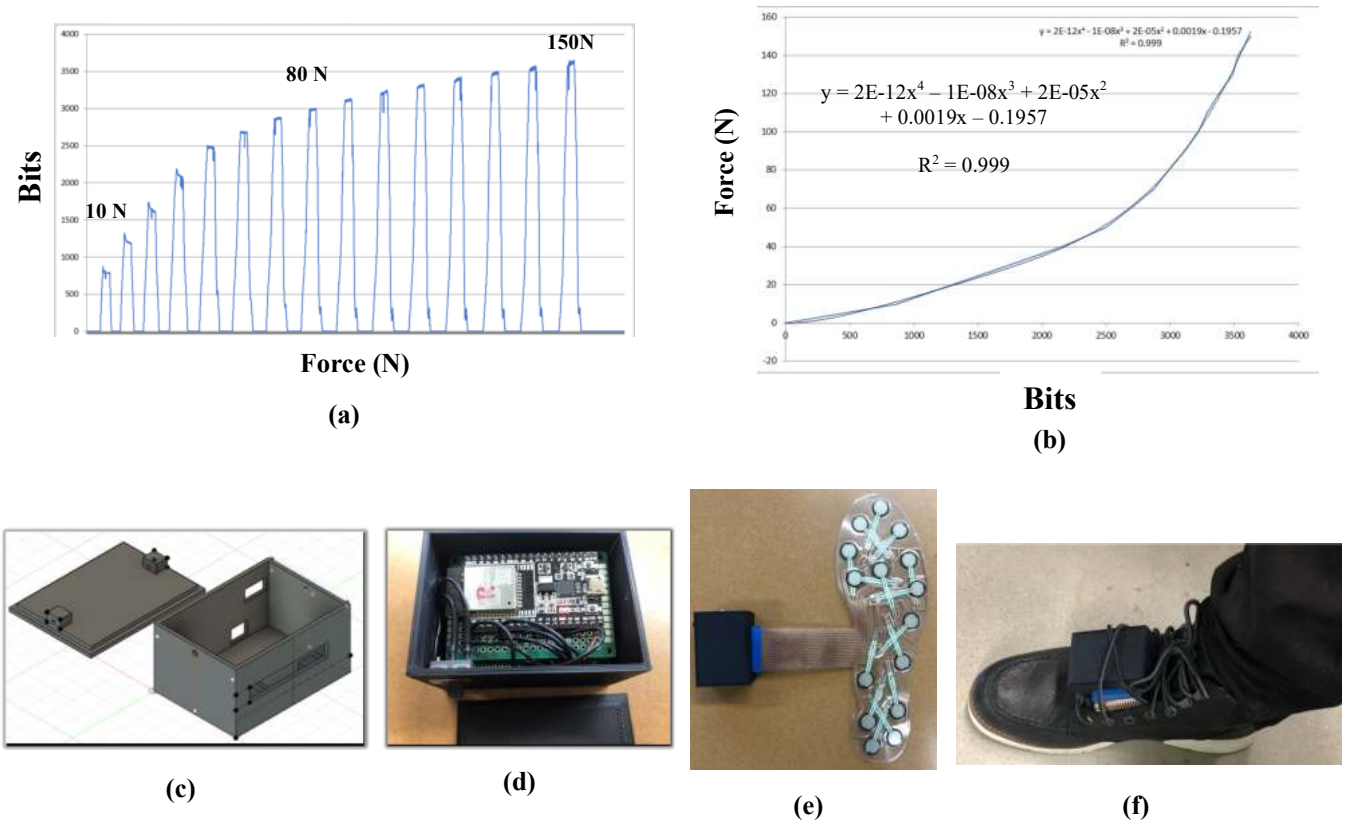

Fig. 2: (a) Test results for force testing between 0 to 150 N with 10 N intervals, (b) retrieved fourth order equation for the force, (c) designed casing model using Autodesk, (d) 3D printed casing holding the microcontrollers and, (e) final product of right insole connected to the circuit.

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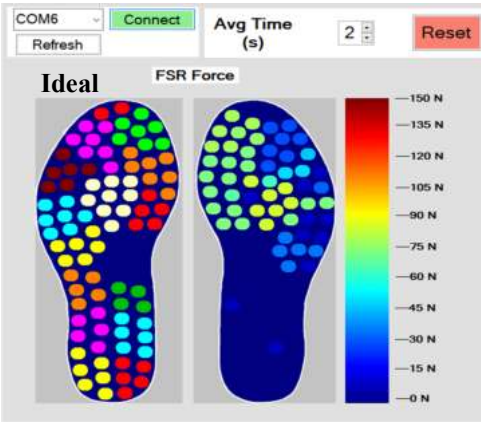
the correct COM port, the application was started which displayed the numerical value of the force and average force for each sensor. The average time to calculate the average force was set from 1 second up to 30 seconds. Similarly, the insole on the right was connected while in use, showing a subject using the insole, leaning forward, and placing pressure towards the front of their foot and toes. The force shows blue at 0 Newtons and goes up red at 130 N and above.

Later, using the smart insole gait analysis was performed. Figure 3(c) represents the gait curve monitoring in different strides such as heel strike, flat foot, mid stance, push off, and toe off. Among these different strides heel strike, foot flat and push off are the key readings that should be considered when analyzing since most pressure is applied during these strides when the subject is in motion or walking. The gait curve clearly shows the main phases of the gait cycle in different strides. Further analysis on these curves can provide insight on various medical diagnosis. For example, if the subject using the smart insole has knee pain, and notices on the initial impact force on the heel is significantly greater than the push off force, we can conclude that the subject is taking too large of a stride. Overall, the fabricated smart insole can be accessed from anywhere without issues regarding to the comfort. Considering its flexible design, it can be fit into shoes of different sizes and shapes and can provide real time pressure map to the user.

IV. Conclusion

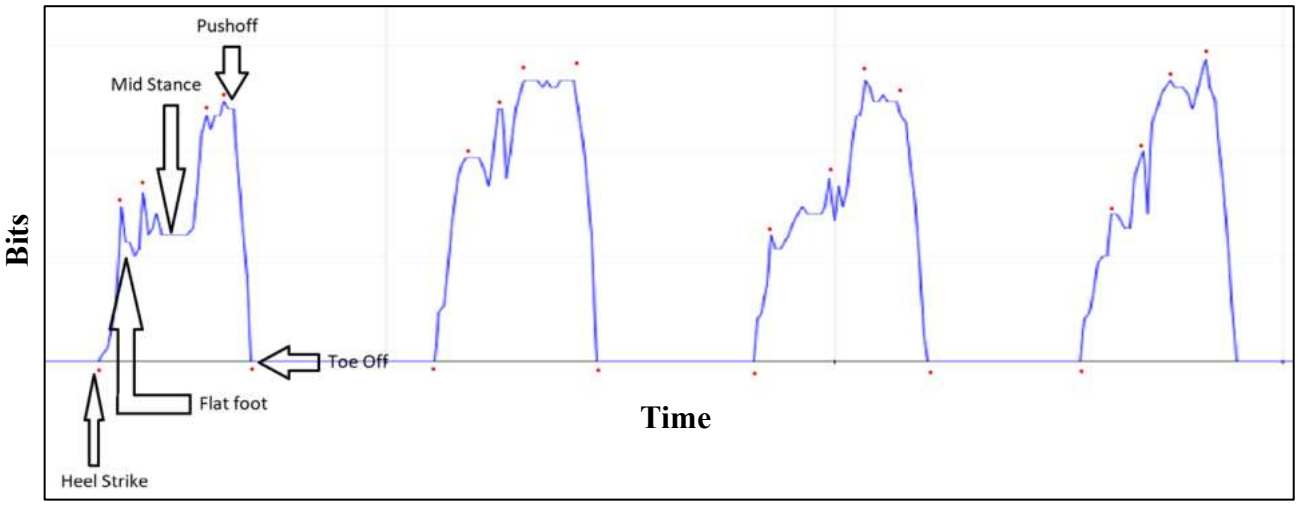
A smart insole for gait monitoring and pressure distribution was successfully fabricated and tested. The fabricated insole was wireless and compatible with compatible with computer applications along with cost-efficient production when compared to traditional gait monitoring devices. Further research will be conducted by developing smartphone application for easy travels as well as introduce different analysis along with introducing Wi-Fi advances for backup storage & cloud capabilities. Additive printing techniques such as screen printing will be investigated for printing insole traces and adding IMU sensors for 3D graph.



(a)

Sensor	Force (N)	Avg (N)
1	0	0
2	50	29.93
3	97	21.2
4	102	38.85
5	0	7.9
6	46	14.16
7	38	25
8	0	0.33
9	0	0
10	0	0.13
11	19	20.15
12	0	23.25
13	0	39.08
14	0	39.78
15	0	48.28

(b)



(c)

Fig. 3: (a) Layout of pressure mapping using visual studio, (b) force measured on respective sensors depending on the subject's posture and, (c) gait curve monitoring representing main phases (heel strike, flat foot, mid stance, push off, and toe off).

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